Simple Statement Of Selection Rule:

Under ordinary conditions spectrum lines corresponding to the transition of an electron from any one state to another are not observed. In emission or absorption spectrum a selection rule is found to be followed which may be derived from Bohr's correspondence principle as well as quantum mechanics.

1. The most simple restriction is on the principal quantum number 'n' which during the transition may change by $\Delta n = (0), 1, 2, 3 \ldots$

2. In any single electron transition quantum number 'l' must change by ± 1 i.e $\Delta l = \pm 1$.

3. Considering the spin orbital interaction we get a new quantum number, known as total atomic quantum number j given by $j = l+s$ or $j = l-s$ which gives rise to a spliting of l levels.
During transition of an electron j values can change only by 0 or ± 1

i.e $\Delta j = 0, \pm 1$

but transition from $j_1 = 0$ to $j_2 = 0$ is not allow

4. Since the total angular momentum j is spatially quantised we get a net quantum no. mj which can have $(2j+1)$ values

from $-j$ to j in unit steps. During a transition the m_j value can change by 0 or ± 1 i.e $\Delta m_j = 0, \pm 1$

If the Sodium atom is excited by some means or other so that the single valance electron is raised to one of the excited state 3^2p, 4^2s, 3^2d, 4^2s .... The electron may return to the normal states 3^2S by making only certain selected jumps.

According to the Selection principle $\Delta l = \pm 1$ the electron in the 3^2d state can not return to 3^2S state directly but it has to come back in two separate jumps 3^2D to 3^2P with the emission of 1st line in the Diffuse series written as $3^2P - 3^2D$ and then from 3^2P to 3^2S giving rise to the 1st line in the principal series. For the time being we are not considering the splitting of l-levels (i.e the quantum no j) which gives rise to fine structure.

If the electron is in 4^2P state it has three possible roots open along which it may return to the normal state. 1st by direct transition from 4^2P to 3^2S. 2nd by three transitions 4^2P to 3^2D, 3^2D to 3^2P and then 3^2P to 3^2S. 3rd by three transitions 4^2P to 4^2S, 4^2S to 3^2P, 3^2P to 3^2S.

There are some exceptions to the selection rule for the quantum number l but the forbidden transitions are so rare that the spectral lines are exceedingly faint.